Marble Solver Derivation

Robotic Exploration Lab

June 30, 2026

1 Problem Setup

We seek to solve the following Quadratic Program with Complementarity Constraints (QPCC):

$$
\begin{aligned}
\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad & \frac{1}{2}x^\top Q x + g^\top x + c \\
\text{s.t.} \quad & Ax + b = 0 \\
& Gx + h \geq 0 \\
& 0 \leq Lx + l \perp Rx + r \geq 0
\end{aligned} \tag{1}
$$

for $Q \succeq 0$, $A \in \mathbb{R}^{m\times n}$, $G \in \mathbb{R}^{p\times n}$, $L, R \in \mathbb{R}^{c\times n}$, $b \in \mathbb{R}^m$, $h \in \mathbb{R}^p$, and $l, r \in \mathbb{R}^c$.

2 Derivation

We introduce the *retraction map* $b_\kappa(x)$ which satisfies:

$$
b_\kappa(x) \cdot b_\kappa(-x) = \kappa \tag{2}
$$

Using (28), we can relax the complementarity constraints in (1) as:

$$
\begin{aligned}
\underset{x\in\mathbb{R}^n, \sigma\in\mathbb{R}^c, d\in\mathbb{R}^p}{\text{minimize}} \quad & \frac{1}{2}x^\top Q x + g^\top x + c - \kappa \mathbf{1}^T \log(d) \\
\text{s.t.} \quad & Ax + b = 0 \\
& Gx + h = d \\
& Lx + l = b_\kappa(\sigma) \\
& Rx + r = b_\kappa(-\sigma)
\end{aligned} \tag{3}
$$

We then introduce the Augmented Lagrangian for (29) as:

$$
\begin{aligned}
\mathcal{L}_\rho(x, \sigma, d; \alpha, \mu, \gamma_L, \gamma_R) = & \frac{1}{2}x^\top Q x + g^\top x + c \\
& + \mu^\top (Gx + h - d) - \kappa \mathbf{1}^T \log(d) \\
& + \alpha^\top (Ax + b) + \frac{\rho}{2} \left\| Ax + b \right\|^2 \\
& + \gamma_L^\top (Lx + l - b_\kappa(\sigma)) + \frac{\rho}{2} \left\| Lx + l - b_\kappa(\sigma) \right\|^2 \\
& + \gamma_R^\top (Rx + r - b_\kappa(-\sigma)) + \frac{\rho}{2} \left\| Rx + r - b_\kappa(-\sigma) \right\|^2
\end{aligned} \tag{4}
$$

Using an Augmented-Lagrangian approach, we can solve (29) by iteratively minimizing (30) over primal variables x, σ, d. We update AL multipliers $\alpha, \gamma_L, \gamma_R$ in the outer loop, while μ is solved as an IPM dual variable in the inner loop.

We first derive stationarity with respect to the original primal variable x. Define the following shorthands:

$$r_A := Ax + b \tag{5}$$
$$r_G := Gx + h - d \tag{6}$$
$$r_L := Lx + l - b_\kappa(\sigma) \tag{7}$$
$$r_R := Rx + r - b_\kappa(-\sigma) \tag{8}$$

We therefore have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial x} = Qx + g + A^\top \alpha + \rho A^\top r_A + G^\top \mu + L^\top \gamma_L + \rho L^\top r_L + R^\top \gamma_R + \rho R^\top r_R = 0 \tag{9}$$
$$= Qx + g + A^\top \lambda + G^\top \mu + L^\top \tau_L + R^\top \tau_R = 0 \tag{10}$$

We introduce the primal-dual shifted multiplier variables:

$$\lambda = \alpha + \rho r_A = \alpha + \rho(Ax + b) \tag{11}$$
$$\tau_L = \gamma_L + \rho r_L = \gamma_L + \rho(Lx + l - b_\kappa(\sigma)) \tag{12}$$
$$\tau_R = \gamma_R + \rho r_R = \gamma_R + \rho(Rx + r - b_\kappa(-\sigma)) \tag{13}$$

For variable d, we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial d} = -\mu - \kappa d^{-1} = 0 \tag{14}$$
$$\iff \mu \odot d = -\kappa \mathbf{1} \tag{15}$$

We can satisfy this relaxed complementarity condition by again using (28) as:

$$d = b_\kappa(v), \quad \mu = -b_\kappa(-v), \quad v \in \mathbb{R}^p \tag{16}$$

And finally for variable σ, we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial \sigma} = -b'_\kappa(\sigma)(\gamma_L + \rho r_L) + b'_\kappa(-\sigma)(\gamma_R + \rho r_R) = 0 \tag{17}$$
$$= -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R = 0 \tag{18}$$

In all, we have the following KKT conditions for (29):

$$r_x := Qx + g + A^\top \lambda + G^\top \mu + L^\top \tau_L + R^\top \tau_R \quad = 0 \tag{19}$$
$$r_\mu := \mu + b_\kappa(-v) \quad = 0 \tag{20}$$
$$r_\sigma := -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R \quad = 0 \tag{21}$$
$$r_\lambda := Ax + b - \rho^{-1}(\lambda - \alpha) \quad = 0 \tag{22}$$
$$r_i := Gx + h - b_\kappa(v) \quad = 0 \tag{23}$$
$$r_{\tau_L} := Lx + l - b_\kappa(\sigma) - \rho^{-1}(\tau_L - \gamma_L) \quad = 0 \tag{24}$$
$$r_{\tau_R} := Rx + r - b_\kappa(-\sigma) - \rho^{-1}(\tau_R - \gamma_R) \quad = 0 \tag{25}$$

We now write the Newton system for the KKT conditions described above. We note that $b''_\kappa(\sigma) = b''_\kappa(-\sigma)$ since $b''_\kappa(x)$ is an even function.

$$\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & -b'_\kappa(-v) & 0 & 0 & I & 0 & 0 \\ 0 & 0 & -b''_\kappa(\sigma)\odot(\tau_L+\tau_R) & 0 & 0 & -b'_\kappa(\sigma) & b'_\kappa(-\sigma) \\ A & 0 & 0 & -\rho^{-1}I & 0 & 0 & 0 \\ G & -b'_\kappa(v) & 0 & 0 & 0 & 0 & 0 \\ L & 0 & -b'_\kappa(\sigma) & 0 & 0 & -\rho^{-1}I & 0 \\ R & 0 & b'_\kappa(-\sigma) & 0 & 0 & 0 & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta\sigma \\ \Delta\lambda \\ \Delta\mu \\ \Delta\tau_L \\ \Delta\tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ r_\mu \\ r_\sigma \\ r_\lambda \\ r_i \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \tag{26}$$

Using the second equation $-b'_\kappa(-v)\Delta v + \Delta\mu = -r_\mu$, we solve for $\Delta\mu$ and get:

$$Q\Delta x + A^\top\Delta\lambda + G^\top b'_\kappa(-v)\Delta v + L^\top\Delta\tau_L + R^\top\Delta\tau_R = -r_x + G^\top r_\mu$$

From this, we subtract $G^\top$ times the fifth equation $G\Delta x - b'_\kappa(v)\Delta v = -r_i$ to get:

$$(Q - G^\top G)\Delta x + A^\top\Delta\lambda + G^\top\Delta v + L^\top\Delta\tau_L + R^\top\Delta\tau_R = -r_x + G^\top(r_\mu + r_i)$$

where we've used the identity $b'_\kappa(-v) + b'_\kappa(v) = I$. The system is now symmetric, and written as:

$$\begin{bmatrix} Q - G^\top G & G^\top & 0 & A^\top & L^\top & R^\top \\ G & -b'_\kappa(v) & 0 & 0 & 0 & 0 \\ 0 & 0 & -b''_\kappa(\sigma)\odot(\tau_L+\tau_R) & 0 & -b'_\kappa(\sigma) & b'_\kappa(-\sigma) \\ A & 0 & 0 & -\rho^{-1}I & 0 & 0 \\ L & 0 & -b'_\kappa(\sigma) & 0 & -\rho^{-1}I & 0 \\ R & 0 & b'_\kappa(-\sigma) & 0 & 0 & -\rho^{-1}I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta\sigma \\ \Delta\lambda \\ \Delta\tau_L \\ \Delta\tau_R \end{bmatrix} = - \begin{bmatrix} r_x - G^\top(r_\mu + r_i) \\ r_\sigma \\ r_\lambda \\ r_i \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \tag{27}$$

3 Derivation

We introduce the *retraction map* $b_\kappa(x)$ which satisfies:

$$b_\kappa(x) \cdot b_\kappa(-x) = \kappa \tag{28}$$

Using (28), we can relax the complementarity constraints in (1) as:

$$\begin{aligned} \underset{x\in\mathbb{R}^n,\sigma\in\mathbb{R}^c,d\in\mathbb{R}^p}{\text{minimize}} \quad & \frac{1}{2}x^\top Qx + g^\top x + c - \kappa\mathbf{1}^T\log(d) \\ \text{s.t.} \quad & Ax + b = 0 \\ & Gx + h = d \\ & Lx + l = b_\kappa(\sigma) \\ & Rx + r = b_\kappa(-\sigma) \end{aligned} \tag{29}$$

We then introduce the Augmented Lagrangian for (29) as:

$$\begin{aligned} \mathcal{L}_\rho(x,\sigma,d;\alpha,\beta,\gamma_L,\gamma_R) =& \frac{1}{2}x^\top Qx + g^\top x + c - \kappa\mathbf{1}^T\log(d) \\ &+ \alpha^\top(Ax+b) + \frac{\rho}{2}\|Ax+b\|^2 \\ &+ \beta^\top(Gx+h-d) + \frac{\rho}{2}\|Gx+h-d\|^2 \\ &+ \gamma_L^\top(Lx+l-b_\kappa(\sigma)) + \frac{\rho}{2}\|Lx+l-b_\kappa(\sigma)\|^2 \\ &+ \gamma_R^\top(Rx+r-b_\kappa(-\sigma)) + \frac{\rho}{2}\|Rx+r-b_\kappa(-\sigma)\|^2 \end{aligned} \tag{30}$$

Using an Augmented-Lagrangian approach, we can solve (29) by iteratively minimizing (30) over primal variables x, σ, d and updating dual variables $\alpha, \beta, \gamma_L, \gamma_R$.

We first derive stationarity with respect to the original primal variable x. Define the following shorthands:

$$r_A := Ax + b \tag{31}$$

$$r_G := Gx + h - d \tag{32}$$

$$r_L := Lx + l - b_\kappa(\sigma) \tag{33}$$

$$r_R := Rx + r - b_\kappa(-\sigma) \tag{34}$$

We therefore have the following stationarity condition:

$$\frac{\partial\mathcal{L}_\rho}{\partial x} = Qx + g + A^\top\alpha + \rho A^\top r_A + G^\top\beta + \rho G^\top r_G + L^\top\gamma_L + \rho L^\top r_L + R^\top\gamma_R + \rho R^\top r_R = 0 \tag{35}$$

$$= Qx + g + A^\top\lambda + G^\top\mu + L^\top\tau_L + R^\top\tau_R = 0 \tag{36}$$

We introduce the primal-dual shifted multiplier variables:

$$\lambda = \alpha + \rho r_A = \alpha + \rho(Ax+b) \tag{37}$$

$$\mu = \beta + \rho r_G = \beta + \rho(Gx+h-d) \tag{38}$$

$$\tau_L = \gamma_L + \rho r_L = \gamma_L + \rho(Lx+l-b_\kappa(\sigma)) \tag{39}$$

$$\tau_R = \gamma_R + \rho r_R = \gamma_R + \rho(Rx+r-b_\kappa(-\sigma)) \tag{40}$$

For variable d, we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial d} = -\beta - \rho r_G - \kappa d^{-1} = 0 \tag{41}$$
$$= -\mu - \kappa d^{-1} = 0 \iff \mu \odot d = -\kappa \mathbf{1} \tag{42}$$

We can satisfy this relaxed complementarity condition by again using (28) as:

$$d = b_\kappa(v), \quad \mu = -b_\kappa(-v), \quad v \in \mathbb{R}^p \tag{43}$$

And finally for variable σ, we have the following stationarity condition:

$$\frac{\partial \mathcal{L}_\rho}{\partial \sigma} = -b'_\kappa(\sigma)(\gamma_L + \rho r_L) + b'_\kappa(-\sigma)(\gamma_R + \rho r_R) = 0 \tag{44}$$
$$= -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R = 0 \tag{45}$$

In all, we have the following KKT conditions for (29). We note that $b''_\kappa(\sigma) = b''_\kappa(-\sigma)$ since $b''_\kappa(x)$ is an even function.

$$r_x := Qx + g + A^\top \lambda + G^\top \mu + L^\top \tau_L + R^\top \tau_R = 0 \tag{46}$$
$$r_v := \mu + b_\kappa(-v) = 0 \tag{47}$$
$$r_\sigma := -b'_\kappa(\sigma)\tau_L + b'_\kappa(-\sigma)\tau_R = 0 \tag{48}$$
$$r_\lambda := Ax + b - \rho^{-1}(\lambda - \alpha) = 0 \tag{49}$$
$$r_\mu := Gx + h - b_\kappa(v) - \rho^{-1}(\mu - \beta) = 0 \tag{50}$$
$$r_{\tau_L} := Lx + l - b_\kappa(\sigma) - \rho^{-1}(\tau_L - \gamma_L) = 0 \tag{51}$$
$$r_{\tau_R} := Rx + r - b_\kappa(-\sigma) - \rho^{-1}(\tau_R - \gamma_R) = 0 \tag{52}$$

To solve this root-finding problem, we first write the asymmetric Newton system as the following, where we note $b''_\kappa(\sigma) = b''_\kappa(-\sigma)$:

$$\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & -b'_\kappa(-v) & 0 & 0 & I & 0 & 0 \\ 0 & 0 & -b''_\kappa(\sigma) \odot (\tau_L + \tau_R) & 0 & 0 & -b'_\kappa(\sigma) & b'_\kappa(-\sigma) \\ A & 0 & 0 & -\rho^{-1} I & 0 & 0 & 0 \\ G & -b'_\kappa(v) & 0 & 0 & -\rho^{-1} I & 0 & 0 \\ L & 0 & -b'_\kappa(\sigma) & 0 & 0 & -\rho^{-1} I & 0 \\ R & 0 & b'_\kappa(-\sigma) & 0 & 0 & 0 & -\rho^{-1} I \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta \sigma \\ \Delta \lambda \\ \Delta \mu \\ \Delta \tau_L \\ \Delta \tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ r_v \\ r_\sigma \\ r_\lambda \\ r_\mu \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \tag{53}$$

Using the second equation $-b'_\kappa(-v)\Delta v + \Delta \mu = -r_v$, we solve for $\Delta \mu$ and get:

$$Q\Delta x + A^\top \Delta \lambda + G^\top b'_\kappa(-v)\Delta v + L^\top \Delta \tau_L + R^\top \Delta \tau_R = -r_x + G^\top r_v$$

We then subtract $G^\top$ times the fifth equation $G\Delta x - b'_\kappa(v)\Delta v - \rho^{-1}\Delta \mu = -r_\mu$ to get:

$$(Q - G^\top G)\Delta x + A^\top \Delta \lambda + G^\top (I + \rho^{-1} b'_\kappa(-v))\Delta v + L^\top \Delta \tau_L + R^\top \Delta \tau_R = -r_x + G^\top (r_v + r_\mu + \rho^{-1} r_v)$$

$$\begin{bmatrix} Q - G^\top G & G^\top (I + \rho^{-1} b'_\kappa(-v)) & 0 & A^\top & G^\top & L^\top & R^\top \\ G & -b'_\kappa(v) & 0 & 0 & -\rho^{-1} I & 0 & 0 \\ 0 & 0 & -b''_\kappa(\sigma) \odot (\tau_L + \tau_R) & 0 & 0 & -b'_\kappa(\sigma) & b'_\kappa(-\sigma) \\ A & 0 & 0 & -\rho^{-1} I & 0 & 0 & 0 \\ L & 0 & -b'_\kappa(\sigma) & 0 & 0 & -\rho^{-1} I & 0 \\ R & 0 & b'_\kappa(-\sigma) & 0 & 0 & 0 & -\rho^{-1} I \end{bmatrix} \tag{54}$$

We symmetrize this system by multiplying the second set of equations by $-b'_\kappa(v)$ to arrive at the following symmetric Newton system:

$$\underbrace{\begin{bmatrix} Q & 0 & 0 & A^\top & G^\top & L^\top & R^\top \\ 0 & b'_\kappa(v) b'_\kappa(-v) & 0 & 0 & -b'_\kappa(v) & 0 & 0 \\ 0 & 0 & -b''_\kappa(\sigma) \odot (\tau_L + \tau_R) & 0 & 0 & -b'_\kappa(\sigma) & b'_\kappa(-\sigma) \\ A & 0 & 0 & -\rho^{-1} I & 0 & 0 & 0 \\ G & -b'_\kappa(v) & 0 & 0 & -\rho^{-1} I & 0 & 0 \\ L & 0 & -b'_\kappa(\sigma) & 0 & 0 & -\rho^{-1} I & 0 \\ R & 0 & b'_\kappa(-\sigma) & 0 & 0 & 0 & -\rho^{-1} I \end{bmatrix}}_{K} \begin{bmatrix} \Delta x \\ \Delta v \\ \Delta \sigma \\ \Delta \lambda \\ \Delta \mu \\ \Delta \tau_L \\ \Delta \tau_R \end{bmatrix} = - \begin{bmatrix} r_x \\ -b'_\kappa(v) r_v \\ r_\sigma \\ r_\lambda \\ r_\mu \\ r_{\tau_L} \\ r_{\tau_R} \end{bmatrix} \tag{55}$$

We can decompose this symmetric KKT system into the following block form:

$$K = \begin{bmatrix} H & J^\top \\ J & -\rho^{-1} I \end{bmatrix} \tag{56}$$

where:

$$H = \begin{bmatrix} Q & 0 & 0 \\ 0 & b'_\kappa(v) b'_\kappa(-v) & 0 \\ 0 & 0 & -b''_\kappa(\sigma) \odot (\tau_L + \tau_R) \end{bmatrix}, \qquad J = \begin{bmatrix} A & 0 & 0 \\ G & -b'_\kappa(v) & 0 \\ L & 0 & -b'_\kappa(\sigma) \\ R & 0 & b'_\kappa(-\sigma) \end{bmatrix} \tag{57}$$

3.1 Inertia Correction

By Haynsworth's inertia additivity, the inertia of the KKT matrix is given by:

$$\mathrm{In}(K) = \mathrm{In}(-\rho^{-1} I) + \mathrm{In}(H + \rho J^\top J) \tag{58}$$

The target inertia for the KKT system is $\mathrm{In}(K) = (n + p + c, m + p + 2c, 0)$; since $\mathrm{In}(-\rho^{-1} I) = (0, m+p+2c, 0)$, we require that $\mathrm{In}(H + \rho J^\top J) = (n+p+c, 0, 0)$, or equivalently that $H + \rho J^\top J \succ 0$. We can ensure this by adding a diagonal regularization term δI to H such that:

$$H + \delta I + \rho J^\top J \succ 0 \tag{59}$$

For further analysis, we write out the matrix $\rho J^\top J$ as:

$$\rho J^\top J = \rho \begin{bmatrix} A^\top A + G^\top G + L^\top L + R^\top R & -G^\top b'_\kappa(v) & -L^\top b'_\kappa(\sigma) + R^\top b'_\kappa(-\sigma) \\ -b'_\kappa(v) G & b'_\kappa(v)^2 & 0 \\ -b'_\kappa(\sigma) L + b'_\kappa(-\sigma) R & 0 & b'_\kappa(\sigma)^2 + b'_\kappa(-\sigma)^2 \end{bmatrix} \tag{60}$$

4 Algorithm